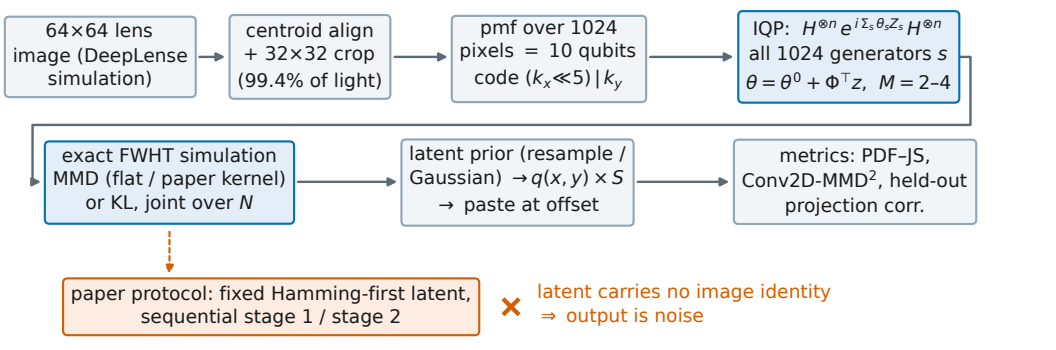
64×64 lens image (DeepLense simulation)
centroid align + 32×32 crop (99.4% of light)
pmf over 1024 pixels = 10 qubits code ($k_x \ll 5$) | k_y
IQP: $H^{\otimes n} e^{i \Sigma_s \theta_s Z_s} H^{\otimes n}$ all 1024 generators s $\theta = \theta^0 + \Phi^\top z$, M = 2–4
exact FWHT simulation MMD (flat / paper kernel) or KL, joint over N
latent prior (resample / Gaussian) $\to q(x, y) \times S$ → paste at offset
metrics: PDF–JS, Conv2D-MMD^2, held-out projection corr.
paper protocol: fixed Hamming-first latent, sequential stage 1 / stage 2
latent carries no image identity
⇒ output is noise